Toward a four-dimensional Yang–Mills construction

Notebook 21: local classical actions through arbitrary refinement

Research working document

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Abstract

We localize the actual homogeneous constrained classical action uniformly through the preceding refinement $M = L^k$. Smooth physical cells and the normalized fine pairing give uniform mixed-insertion bounds for the concrete Wilson and corrected constraint vertices. A contraction in a rooted support algebra then constructs the minimizer and multiplier as convergent families of functions of specified background cells. The action-envelope identity gives strictly supported scalar coefficients; their Hessians are individually symmetric local precision pieces. The algebra admits an arbitrary fixed support-cardinality weight, at the cost of a smaller uniform background ball. Substitution into the exact fixed-L source chart gives the localized classical remainder and physical precision needed by the overlap integration theorem. This extends the bare small-field integral of Notebook 20 to one integration of the classical reference action at any earlier classical depth. Generated quantum interactions and large-field histories still require an induction theorem.

The distinction between the two kinds of depth is explicit: S_M is obtained by classical minimization, while a full quantum effective action also contains the corrections from every earlier fluctuation integral. The theorem here controls the former and one subsequent small-field integration. All strict support claims concern the original independent current-background variables.

1 Concrete classical vertices uniform through refinement

The fine spacing is $\eta = M^{-1}$, with $M = L^j$, fixed odd L, dimension four and current spacing one. Fine pairings use η^4 times counting measure; current pairings use counting measure. All constants below are uniform in M and compatible periodic volume. The conclusion is a set of concrete inputs for a rooted support expansion, not an assumption that one weighted response estimate already gives a sum over all input insertions.

1.1 Complete physical field spaces and known bounds

Fix $0 < \alpha < 1$ and use complete fine-field spaces

$$\begin{aligned} X_\eta = \mathcal{N}_\alpha : \quad & \|A\|_X = \|A\|_\infty + \|\nabla_\eta A\|_\infty + [\nabla_\eta A]_\alpha, \\ & Y_\eta = C^\alpha \text{ of fine jet coefficients}, \\ & Z_\eta = \ell^\infty \text{ of fine coefficients}, \qquad E = \ell^\infty \text{ of current bond fields}. \end{aligned} \tag{1}$$

Local Hölder norms use physical distances and fixed cell collars. Color and orientation spaces have fixed finite dimension. The plaquette jet $S = (I, \nabla_\eta)$, with the needed finite component selections, maps X_η to Y_η uniformly. No sharp cell indicator is differentiated.

The proved Wilson-force module gives

$$\mathcal{S}_\eta(A) = \eta^4 \sum_p \mathcal{L}_\eta((SA)_p), \qquad N(A) = S^T f_\eta(SA),$$

where $\mathcal{L}_\eta$ is uniformly analytic in the jet, its quadratic part is the Maxwell density, and f_η starts at degree two. The full corrected constraint of Notebook13 is

$$\mathcal{F}(A) = QA + \mathcal{R}(A), \qquad Q = Q_M^{\mathrm{st}}.$$

It is analytic on a refinement-uniform supremum ball and has a fixed physical dependency radius. Notebook15 also proves uniform holomorphy of $D\mathcal{R}(A)^T$ in the physical bounded-row operator norm, with norm $O(\|A\|_\infty)$. The transpose includes the physical fine measure.

The weighted complete-field bounds for $H, \mathcal{C}$ and their differentiated compositions are recorded explicitly in Section 1.5. They include the equal-exponent constrained Schauder theorem of Notebook15. The scalar action of the ordinary homogeneous minimizer is exactly the action at the corrected fixed-slice minimizer by Notebook15's gauge transfer. This does not identify the two constraint derivatives.

The underlying corrected one-step coordinate map is specified at Raw I, lines 24320–24325 and 24344–24369, with the ordinary/corrected bridge at 47076–47087. Its physical scaling is $(Lh)^{-1}F_L(hA)$. The Wilson density and pairing use the conventions already matched in Notebooks14–15. No additional claim about the full original orbit domains is made here.

1.2 Smooth cell spaces without an extension assumption

Choose a smooth partition of unity $\{\chi_i\}$ indexed by unit current cells, with support in fixed enlarged cells S_i, uniformly bounded derivatives through order three, and uniformly finite overlap. For example, divide a nonnegative compactly supported smooth bump by the sum of its integer translates. Periodize on compatible tori, counting repeated supports with multiplicity.

For each fine field type let E_i be the Banach subspace of *complete* fields supported in S_i, with its global X_η, Y_η or Z_η norm. Write I_i for inclusion and $P_i u = \chi_i u$. For current fields use coordinate restriction and insertion in one current cell. Then

$$u = \sum_i I_i P_i u, \qquad \|P_i u\|_{E_i} \le C \, \|u\|_{\text{fixed collar of } i}. \tag{2}$$

The reconstruction

$$\mathscr{R}((u_i)) = \sum_i I_i u_i$$

is bounded from the block supremum space to the complete global space, uniformly in η and volume. In particular, $\mathscr{R}P = I$ and $P\mathscr{R}$ is a bounded projection. Arbitrary restrictions on a cell are not extended by an unproved extension theorem; the spaces E_i already consist of supported complete fields.

For the only delicate norm X_η, the exact product rule is

$$\nabla_\eta(\chi A)(x) = \chi(x + \eta e)\nabla_\eta A(x) + (\nabla_\eta \chi)(x) A(x). \tag{3}$$

The sampled smooth functions and their first differences have uniform physical C^α norms. Telescoping controls the local C^α norm of A by $\|A\|_X$ without an inverse mesh factor. The Hölder product inequality proves (2). Finite overlap, also for pairs of points in fixed collars, proves the reconstruction bound. The arguments for Y_η and Z_η are simpler.

1.3 A summed mixed-insertion vertex bound

For a finite set of current cell labels S, let $\tau(S)$ be the minimum length of a connected current-lattice graph containing S, allowing unmarked auxiliary vertices. This is graph Steiner length. It is not the minimum-spanning-tree length on the terminals alone.

For a local analytic map F, expand after reconstruction:

$$P_i F(\mathscr{R}u) = \sum_{n\geq 0} \sum_{j_1,\dots,j_n} V_{n;i;j_1,\dots,j_n}[u_{j_1},\dots,u_{j_n}]. \tag{4}$$

The coefficients contain $1/n!$. A fixed finite direct sum handles several field types. Define

$$v_n(\kappa) = \sup_i \sum_{j_1,\dots,j_n} e^{\kappa\tau(\{i,j_1,\dots,j_n\})} \left\| V_{n;i;j_1,\dots,j_n} \right\|. \tag{5}$$

The coefficient norm is the multilinear operator norm of the cell Banach spaces. All input cell labels are summed.

Lemma 1.1 (Finite-range analytic vertices)**.** *Suppose each output reads only labels within distance R, at most D_R labels, and F is analytic and uniformly bounded by M_F on a global ball of radius r_F. If $\|\mathscr{R}\| \leq C_P$, then*

$$v_n(\kappa) \leq M'_F \left(\frac{eC_P D_R e^{\kappa R}}{r_F} \right)^n, \tag{6}$$

with a fixed output-localization factor in M'_F. If F starts at degree two, the generating majorant is analytic and quadratic at zero.

Proof. For normalized directions, multivariable Cauchy with radii $r_F/(nC_P)$ bounds the derivative by $M'_F(nC_P/r_F)^n$. Division by $n!$ and $n^n/n! \leq e^n$ gives the coefficient bound. There are at most D_R^n nonzero ordered tuples. Joining the root to each input by a shortest path costs at most nR, proving (6). This argument never counts individual fine coordinates. On a smaller analytic margin it also works at a nonzero fixed background, though the constant and linear terms then need not vanish. □

Rooting at an input and summing the output and other inputs has the same bound up to fixed factors, since the possible outputs lie in a fixed neighborhood of that input. For support-family substitution the exact geometric bound is

$$\tau\left(\{i\} \cup \bigcup_r X_r\right) \leq \tau(\{i, j_1, \dots, j_n\}) + \sum_r \tau(\{j_r\} \cup X_r). \tag{7}$$

Join the graphs at their roots. The j_r remain allowed as unmarked auxiliary vertices when they are no longer terminals. This constant-one inequality can fail for metric minimum-spanning-tree length. If that convention is denoted by ℓ_{MST}, a doubled traversal and shortcut give

$$\ell_{\mathrm{MST}}(S) \leq 2\tau(S). \tag{8}$$

Thus a final weight $e^{\mu\ell_{\mathrm{MST}}}$ is covered by a Steiner exponent at least 2μ. That spatial reserve must be retained at the polymer interface.

Arbitrary fixed cardinality reserve. The concrete coefficients here are expanded at the fixed flat background. They introduce no extra variable support. For any prescribed $a \geq 0$,

$$e^{a|\cup_r X_r|} \leq \prod_r e^{a|X_r|}. \tag{9}$$

Consequently adding $e^{a|X|}$ to the rooted support norm does not change the nonlinear vertex majorants. Linear operators change the root, not the variable support. Only the initial source B_j, with support $\{j\}$, costs e^a. The source radius may shrink with a while remaining uniform in refinement and volume. This can pay a fixed support-cardinality factor in a later polymer norm; small spatial decay alone would not pay that factor. Hiding a background-dependent minimizer in the Taylor coefficients would require a new support argument and is not done here.

1.4 The actual nonlinear vertices

The Wilson jet coefficient. For a positive plaquette use $z = (a, b, c, e) = (A_\mu, A_\nu, \nabla_\mu A_\nu, \nabla_\nu A_\mu)$ at its base point. The exact expressions are

$$\begin{aligned} P_\eta(z) &= e^{i\eta a} e^{i\eta b + i\eta^2 c} e^{-i\eta a - i\eta^2 e} e^{-i\eta b}, \\ \mathcal{L}_\eta(z) &= -\frac{1}{2}\tau \left[\left(\frac{P_\eta(z) - I}{\eta^2} \right)^2 P_\eta(z)^{-1} \right]. \end{aligned} \tag{10}$$

The singularity at $\eta = 0$ is removed before differentiating vertices: $P_0 = I$ and $\partial_\eta P_\eta|_0 = 0$. Joint analyticity on a fixed jet ball gives uniform coefficients. The bounded jet map and the C^α product estimate make

$$F_\eta^{\mathrm{nl}}(A) = f_\eta(SA) : X_\eta \longrightarrow Y_\eta$$

analytic on a uniform ball and quadratic at zero. Each plaquette output reads a bounded physical neighborhood. Lemma 1.1 therefore proves its full mixed-insertion bound uniformly in η. The transpose S^T stays outside this vertex. A uniform bounded-field estimate for the unpropagated fine force $S^T F_\eta^{\mathrm{nl}}$ is neither used nor claimed.

The Wilson energy density. Set

$$\mathcal{E}_{i,\eta}(A) = \eta^4 \sum_{p>0} \chi_i(x_p) \mathcal{L}_\eta((SA)_p). \tag{11}$$

Then $\mathcal{S}_\eta(A) = \sum_i \mathcal{E}_{i,\eta}(A)$ exactly. Although a current cell contains $O(\eta^{-4})$ fine plaquettes,

$$\eta^4 \sum_{p>0} |\chi_i(x_p)| \le C_d. \tag{12}$$

The fine differences are already bounded in the jet norm. Thus every cell energy is a uniformly bounded analytic scalar map on a fixed $\mathcal{N}_\alpha$ ball, with fixed current dependency range. Its coefficients satisfy (6). Replacing $\mathcal{L}_\eta$ by its remainder after the quadratic Maxwell density gives the same conclusion starting at degree three. Dropping η^4, or treating all fine differences as unrelated bounded coordinates, would lose this estimate. The ordinary constrained scalar action is the same sum at the corrected minimizer by the established gauge transfer.

The full corrected constraint. At input spacing h and output spacing Lh,

$$\begin{aligned} \mathcal{F}_h(A) &= (Lh)^{-1} F_L(hA) = Q_h A + \mathcal{R}_h(A), \\ \|\mathcal{R}_h(A)\|_\infty &\le Ch \|A\|_\infty^2, \qquad \|D\mathcal{R}_h(A)\|_{\infty\to\infty} \le Dh \|A\|_\infty. \end{aligned} \tag{13}$$

For $h_t = L^{t-j}$, $\sum_{t<j} h_t \le (L-1)^{-1}$. The proved Riccati bound keeps every intermediate physical field in a common ball. Hence the composed remainder $\mathcal{R}$ is uniformly analytic, quadratic, and has physical dependency radius at most $r_L \sum_t h_t \le R_*$. It follows that $\mathcal{R} : X_\eta \to E$ satisfies Lemma 1.1. The scale sum is already included in this uniform bound; higher coefficients are obtained by Cauchy in that ball, not by an uncontrolled count of multilevel diagrams.

The physical-adjoint constraint vertex. Define

$$\mathcal{V}_\eta(A, m) = D\mathcal{R}(A)^T m : X_\eta \times E \longrightarrow Z_\eta. \tag{14}$$

Ordinary holomorphy of $\mathcal{R}$ alone would not control this physical transpose. The proved row estimate does: at one step,

$$\left\|D\mathcal{R}_h(A)^T\right\|_{\infty\to\infty} \le D_* h \left\|A\right\|_\infty, \qquad D_* = N_L K_0 L^{d-1}.$$

The factor L^d in the physical transpose is included here. Both Q_h and Q_h^T contract the bounded norm. Transposing the composition and using the intermediate field bound proves

$$\left\|D\mathcal{R}(A)^T\right\|_{\infty\to\infty} \le \exp\left(\frac{2D_* \left\|A\right\|_\infty}{L-1}\right) - 1. \tag{15}$$

The map is holomorphic into this physical absolute-row operator space. Thus (14) is jointly analytic and $O(\|A\|_\infty \|m\|_\infty)$. Its output at a fine bond reads coarse m within R_*, and background A within $2R_*$, because each relevant coarse equation contains that bond and reads its own neighborhood. Lemma 1.1 applies with total degree starting at two. Cauchy is taken in the physical-row operator norm; it is not applied entrywise and followed by a count of fine coefficients. The output is only required to be bounded: it enters $\mathcal{C}$ or H^T, and no derivative of m is taken.

1.5 Summable linear blocks, including the derivative kernels

The established complete-field estimates, for every positive weight with $w(i)/w(j) \le e^{\kappa_0 d(i,j)}$, are

$$\begin{aligned} &H : E_w \to X_{\eta,w}, \qquad \mathcal{C} : Z_{\eta,w} \to X_{\eta,w}, \qquad \mathcal{C}S^T : Y_{\eta,w} \to X_{\eta,w}, \\ &H^T : Z_{\eta,w} \to E_w, \qquad (SH)^T : Z_{\eta,w} \to E_w, \qquad q : E_w \to E_w. \end{aligned} \tag{16}$$

Restricting the fifth input to Y_η is sufficient. The third estimate has the same α at input and output; a positive Hölder loss would not suffice. These are the constrained Schauder and extension bounds already used in the weighted response proof. They do not require the gauge projection Π to act on arbitrary bounded fields.

For any such T, form $T_{ij} = P_i T I_j$. Given a supported input u_j, use $w_j(n) = e^{\kappa_0 d(n,j)}$. Its weighted norm is bounded by its unweighted norm times a fixed collar factor. The complete weighted estimate gives

$$\|T_{ij}\| \le C_T e^{-\kappa_0 d(i,j)}. \tag{17}$$

This is an operator norm between cell Banach spaces, not an absolute integral estimate for a singular differentiated fine-coordinate kernel. In particular its diagonal block is controlled by the equal-exponent Schauder bound. For every $0 < \kappa < \kappa_0$, lattice growth then gives

$$\max\left\{\sup_i \sum_j e^{\kappa d(i,j)} \|T_{ij}\|, \sup_j \sum_i e^{\kappa d(i,j)} \|T_{ij}\|\right\} \le C_{T,\kappa}. \tag{18}$$

Both sums follow from the common envelope, without any self-adjointness assumption in these Banach norms. Periodic-distance sums have the same uniform upper bound.

The decomposition is exact; for example,

$$\mathcal{C}S^T f = \sum_j \mathcal{C}S^T(\chi_j f). \tag{19}$$

Every $\chi_j f$ is a complete C^α field. Smooth cutoff derivatives are included in $S^T(\chi_j f)$ and controlled by the same estimate; they cancel in the sum. No sharp splitting is used. The maps H^T and $(SH)^T$ likewise retain their physical transpose normalization. Output first differences and their Hölder seminorms are already in X_η, while source derivatives occur only through $\mathcal{C}S^T$ and $(SH)^T$. Separate absolute fine-coordinate derivative bounds are unnecessary for (5).

1.6 An exact simultaneous fixed point using these vertices

The corrected constraint is $\mathcal{F}(A) = QA + \mathcal{R}(A)$. Let m be the multiplier of Notebook15 and write $\mathcal{V}_\eta(A, m) = D\mathcal{R}(A)^T m$. The exact stationarity equations are equivalent near zero to

$$\begin{aligned} A &= HB - H\mathcal{R}(A) - \mathcal{C}S^T f_\eta(SA) + \mathcal{C}\mathcal{V}_\eta(A, m), \\ m &= qB - q\mathcal{R}(A) + (SH)^T f_\eta(SA) - H^T \mathcal{V}_\eta(A, m). \end{aligned} \tag{20}$$

Indeed the second equation rearranges to

$$(I + H^T D\mathcal{R}(A)^T)m = q[B - \mathcal{R}(A)] + H^T N(A),$$

and the first is $A = H[B - \mathcal{R}(A)] + \xi$ with $\xi = -\mathcal{C}N(A) + \mathcal{C}D\mathcal{R}(A)^T m$. These are the proved constraint chart and reduced stationarity identities. Conversely the exact constraints of $\mathcal{C}$ put ξ in the fixed flat slice, and the small multiplier inverse recovers the earlier formulation. No extra small critical points are introduced.

All nonlinear primitives in (20) start at degree two: $\mathcal{R}(A)$, $f_\eta(SA)$, and $D\mathcal{R}(A)^T m$. Their mixed-insertion majorants are analytic and uniform by the preceding proof, while their linear propagators satisfy (18). Apply P to the equations and reconstruct with $\mathscr{R}$. A fixed point in compatible cell components reconstructs exactly to (20); the overlap constants are absorbed by reducing the small radius. This supplies the concrete inputs for the separate rooted support-family fixed-point theorem, including any fixed cardinality reserve. Tree summation, canonical grouping of supports and the final scalar-action expansion remain conclusions of that theorem, not substitutes for its proof.

1.7 An obstruction to using the full ordinary adjoint

The full ordinary rooted average is analytic from small complete physical connections to bounded current data, with fixed physical dependency range, by the ordinary trajectory theorem. It therefore has $X_\eta \to E$ vertex bounds. Its physical adjoint on arbitrary bounded current fields is different and is not uniformly bounded through refinement.

Let Q_{ord} be its flat derivative in physical potentials. Exact ordinary gauge covariance gives

$$Q_{\mathrm{ord}} D_\eta \lambda = D_1(\lambda|_{\text{final centers}}), \tag{21}$$

with the same sign on both sides for either consistent gauge convention. On a sufficiently large torus, take a unit color vector λ at one final-center fine site and zero at every other fine site. Its fine gradient is nonzero on $2d$ bonds, each of magnitude η^{-1}, so

$$\|D_\eta \lambda\|_{2,\eta} = \sqrt{2d}\,\eta^{d/2-1}, \qquad \|D_\eta \lambda\|_{1,\eta} = 2d\eta^{d-1}. \tag{22}$$

The current gradient in (21) has magnitude one on $2d$ bonds. Therefore

$$\|Q_{\mathrm{ord}}\|_{L^2_\eta \to L^2_1} \geq \eta^{1-d/2}. \tag{23}$$

Choose m to equal that current gradient, so $\|m\|_\infty = 1$ and $\langle Q_{\mathrm{ord}} D_\eta \lambda, m\rangle_1 = 2d$. Physical transposition and (22) imply

$$\left\|Q_{\mathrm{ord}}^T\right\|_{\infty \to \infty} \geq \eta^{1-d}. \tag{24}$$

In dimension four these are η^{-1} and η^{-3}. The complete $\mathcal{N}_\alpha$ norm of the test gradient is large, so there is no contradiction with $Q_{\mathrm{ord}} : X_\eta \to E$ boundedness.

The corrected flat derivative instead satisfies $Q_M^{\mathrm{st}} D_\eta = D_1 Q_0$ with the scalar block mean, and its physical transpose contracts the bounded norm. This is why (20) uses the corrected map. Possible cancellations in a specially projected ordinary adjoint are not assumed. The ordinary current-step pivot chart of Notebook19 is unaffected: that ratio is the fixed L, so its constants may depend on L. The obstruction concerns unbounded total ratio M.

1.8 Scope of the concrete input

The proved inputs are uniformly analytic finite-current-range Wilson jet and energy vertices, corrected-constraint and physical-adjoint constraint vertices, summed mixed-insertion norms, and exponentially summable linear cell blocks including the derivatives used in the simultaneous equations. The one-weight theorem is used only to form linear block envelopes; the nonlinear coefficient proof really sums all input labels.

No raw fine-coordinate absolute Taylor norm without physical normalization or regularity is asserted. No ordinary rooted adjoint bound, independent-current source extension, mixed-history theorem or quantum integration result is inferred. Backgrounds remain in the homogeneous small global potential branch. The scalar gauge transfer, rather than a false identification of ordinary and corrected derivatives, matches its classical action to the ordinary branch.

2 Strict support localization of the homogeneous classical action

Fix odd L, dimension four, $M = L^j$, $\eta = M^{-1}$ and $0 < \alpha < 1$. Work first on compatible finite periodic regulators. Constants below are independent of M and volume. They may depend on L, color, α, the requested positive spatial exponent and an arbitrarily prescribed finite support-cardinality reserve. The product background radius may shrink with these fixed choices. Fine transposes retain the physical measure η^4.

The concrete cell-vertex estimates preceding this section supply the exact simultaneous equations

$$\begin{aligned} A &= HB - HR(A) - \mathcal{C}S^T f_\eta(SA) + \mathcal{C}V(A,m), \\ m &= qB - qR(A) + (SH)^T f_\eta(SA) - H^T V(A,m), \qquad V(A,m) = DR(A)^T m. \end{aligned} \tag{25}$$

Here $\mathcal{F}(A) = QA + R(A)$ is the full corrected fixed-reference constraint, H is the matched flat minimizing extension, $\mathcal{C}$ the flat constrained inverse, $q = q_M$, and S the local first-difference jet. All three nonlinear primitives start at total degree two. Rearranging the second equation gives the multiplier inverse of the constrained stationary equation; the first then gives its exact constraint chart. Thus (25) is equivalent to the previously proved small classical branch. It uses the physical adjoint of the corrected constraint, not the nonuniform adjoint of the full ordinary rooted average. The scalar action equality supplied by the ordinary-constraint gauge transfer will be used after solving these equations.

2.1 Geometry and the support-family norm

Let d be current-cell graph distance. For a nonempty finite set X, let $\tau(X)$ be the minimum number of edges in a connected graph containing X, allowing unmarked intermediate vertices. This graph Steiner length differs from the metric minimum spanning-tree length ℓ used in the Gaussian expansion, but

$$\tau(X) \le \ell(X) \le 2\tau(X), \qquad |X| \le \tau(X) + 1. \tag{26}$$

The second upper bound follows by traversing a connected minimizing tree and shortcutting its tour in the graph metric. The crucial rooted inequalities are

$$\begin{aligned} \tau(\{i\} \cup X) &\le d(i,j) + \tau(\{j\} \cup X), \\ \tau\left(\{i\} \cup \bigcup_{r=1}^{n} X_r\right) &\le \tau(\{i, j_1, \dots, j_n\}) + \sum_{r=1}^{n} \tau(\{j_r\} \cup X_r). \end{aligned} \tag{27}$$

The auxiliary roots j_r need not be final support labels. This is why the support algebra uses τ: the analogous inequality for metric spanning trees with deleted auxiliary vertices need not hold. All these statements remain valid on periodic graphs.

Use the smooth fixed-physical-scale partition of unity and the complete cell spaces from the concrete vertex construction. Their field types are $\mathcal{N}_\alpha$, C^α, bounded fine fields, and fixed-dimensional current fields. Localization P_i and reconstruction $\mathcal{R}$ are uniformly bounded, and $\mathcal{R}P = I$. In particular, input pieces to $\mathcal{C}S^T$ remain complete C^α fields; no sharp cut is differentiated. For each relevant linear propagator T, the proved weighted complete-field estimate gives

$$\|T_{ij}\| \le C_T e^{-\kappa_0 d(i,j)}, \qquad L_T(\kappa) := \sup_i \sum_j e^{\kappa d(i,j)} \|T_{ij}\| < \infty \quad (0 < \kappa < \kappa_0). \tag{28}$$

The diagonal block of $\mathcal{C}S^T$ is controlled by the proved equal-exponent Schauder estimate, rather than an absolute singular-kernel integral. These block estimates are uniform in refinement and volume.

On the original current-background product ball of radius b, a support family is a collection $u_{i;X}(B|_X)$ of bounded holomorphic functions with values in the cell space at i. The support X is nonempty and records original background cells. For $a \ge 0$ put

$$\|u\|_{\kappa,a;b} := \sup_i \sum_X e^{\kappa\tau(\{i\}\cup X)+a|X|} \sup_{\|B|_X\|_\infty < b} \|u_{i;X}(B|_X)\|. \tag{29}$$

For several field types, take their maximum in each coefficient norm *before* the support sum: the joint family norm is $\sup_i \sum_X e^{\kappa\tau(\{i\}\cup X)+a|X|} \max_t \|u^t_{i;X}\|$, with the displayed background supremum understood. It is not the maximum of the separate type-family norms. Taylor operator norms below use this same joint cell space with its maximum norm. A bound for a joint linear source may be taken as the sum of its component block bounds. These are fixed finite sums. This is a Banach space. Terms with the same support are added in that slot; evaluation $\sum_X u_{i;X}(B|_X)$ converges absolutely. All coefficients remain functions of the original variables. No nonlocal background-dependent minimizer is hidden in their definition.

Lemma 2.1 (Support substitution)**.** *A linear propagator obeying (28) has norm at most $L_T(\kappa)$ in (29). Suppose a nonlinear cell map has Taylor coefficients, including $1/n!$, whose mixed-insertion norms obey*

$$\begin{aligned} v_n &:= \sup_i \sum_{j_1,\dots,j_n} e^{\kappa\tau(\{i,j_1,\dots,j_n\})} \|N_{n;i;j_1,\dots,j_n}\|, \\ \Phi(t) &:= \sum_{n\ge 2} v_n t^n < \infty \quad (0 \le t < t_0). \end{aligned} \tag{30}$$

Assign a substituted term the union of its input supports. Then

$$\begin{aligned} \|N(u)\|_{\kappa,a;b} &\le \Phi(\|u\|_{\kappa,a;b}), \\ \|N(u) - N(v)\|_{\kappa,a;b} &\le \Phi'(r)\|u - v\|_{\kappa,a;b} \quad (\|u\|, \|v\| \le r < t_0). \end{aligned} \tag{31}$$

The majorant is independent of the prescribed cardinality reserve a.

Proof. For the linear assertion attach a path from the input root j to the output root i and use the first inequality in (27). For a nonlinear term use the second inequality and $|\bigcup_r X_r| \le \sum_r |X_r|$. The absolute sum over all supports and ordered input-cell slots then factorizes into the product of their family norms, leaving v_n. Summing over n proves the first bound. Telescoping an n-linear expression gives the factor nr^{n-1} and proves the second. Repeated supports, repeated labels and overlapping supports are all included in these sums. A Taylor coefficient at the fixed flat background introduces no additional variable support, so it adds no cardinality cost. □

The concrete finite-range analytic vertices have

$$v_n \le M_1 \left(\frac{eC_P D_R e^{\kappa R}}{r_0} \right)^n. \tag{32}$$

Here R is a uniform current-cell dependency radius, D_R the fixed number of admissible input cells in one slot, C_P the reconstruction norm, and r_0 a uniform analytic radius. Indeed multivariable Cauchy with radius $r_0/(nC_P)$ bounds the derivative by $M_1(nC_P/r_0)^n$; division by $n!$ and $n^n/n! \le e^n$ bounds its Taylor coefficient. There are at most D_R^n nonzero ordered cell tuples, with a star of length at most nR through the output. This counts current cells, not M^4 independent fine coordinates. For $V(A,m)$ it uses the previously proved holomorphy in the physical absolute-row norm of $DR(A)^T$; ordinary bounded-field holomorphy of R alone would not imply that input.

2.2 A convergent support-valued stationary solution

Theorem 2.2 (Uniform strict-support fixed point)**.** *For every fixed finite $a \ge 0$ and sufficiently small positive κ, there is $b_* > 0$, independent of M and volume, such that the solution of (25) has support families satisfying*

$$\begin{aligned} \|(A,m)\|_{\kappa,a;b} &\le C_a b, \\ \|(A-HB, m-qB)\|_{\kappa,a;b} &\le C_a b^2 \qquad (0 < b \le b_*). \end{aligned} \tag{33}$$

Every coefficient of the second line has Taylor degree at least two.

Proof. Apply the preceding lemma to the three nonlinear primitives in (25) and then their linear propagators. Their combined positive coefficient majorant can be bounded coefficientwise by

$$\Phi(t) \le \frac{v(ct)^2}{1-ct}, \qquad \Phi'(t) \le 8vc^2 t \quad (ct \le 1/2), \tag{34}$$

after increasing fixed v, c. The singleton source $B_{i;\{i\}} = B_i$ has norm $e^a b$. If L_0 bounds the linear map $B \mapsto (HB, qB)$, choose

$$r_* \le \min\{(2c)^{-1}, (16vc^2)^{-1}\}, \qquad b_* \le \frac{r_*}{2L_0 e^a}. \tag{35}$$

Also shrink r_* so that evaluation and reconstruction stay in the previously established classical branch's uniqueness ball; their norms are uniformly bounded. The support map is a contraction with constant at most $1/2$ on the ball of radius $2L_0 e^a b$. This proves the first estimate. Applying the quadratic majorant to its solution proves the second. Degree induction in Picard iteration proves the vanishing orders coefficientwise.

Evaluation is continuous and respects every absolutely convergent substitution. The evaluated solution therefore solves the exact equations (25). The compatible cell image is preserved by their right-hand side, and uniqueness in the common small domain identifies the result with the proved classical minimizer and multiplier. Picard iteration also defines an absolutely convergent rooted-tree expansion: local Taylor coefficients are vertices, flat cell propagators are edges, and the leaves carry the original B_i. All depths, orders and repeated supports are already summed by the Banach contraction. No additional factor counting the number of RG levels occurs. □

2.3 Scalar action and symmetric support Hessians

The established envelope identity is

$$DS_M(B) = m(B), \qquad S_M(0) = 0, \qquad D^2 S_M(0) = q_M. \tag{36}$$

The exact gauge transfer identifies this scalar with the ordinary homogeneous constrained action. Write $n = m - qB$. On a finite regulator,

$$S_M(B) - \tfrac{1}{2}\langle B, qB\rangle = \int_0^1 \sum_i \langle B_i, n_i(tB)\rangle \, dt. \tag{37}$$

The current pairing has unit cell volume and fixed fiber dimension. Define strictly supported scalar coefficients by

$$W_Y(B|_Y) = \sum_{\{i\}\cup X=Y} \int_0^1 \langle B_i, n_{i;X}(tB|_X)\rangle \, dt. \tag{38}$$

They are holomorphic and begin at order three.

Proposition 2.3 (Strict scalar action localization)**.** *For $0 < \mu < \kappa$ and every prescribed finite $a \geq 0$, on a fixed small product ball,*

$$S_M(B) = \tfrac{1}{2}\langle B, q_M B\rangle + \sum_{Y\neq\varnothing} W_Y(B|_Y), \tag{39}$$

$$\sup_b \sum_{Y\ni b} e^{\mu\tau(Y)+a|Y|}\|W_Y\|_r \leq C_{\mu,a} r^3. \tag{40}$$

Here $\|W_Y\|_r$ is the supremum on the complex product ball of radius r. Each Hessian $B_Y = D^2 W_Y$, extended by zero outside cell rows and columns in Y, is individually complex symmetric and depends only on $B|_Y$. With a fixed cardinality spare $\epsilon > 0$,

$$\sup_i \sum_{Y\ni i} \sum_{j\in Y} e^{\mu\tau(Y)+(a-\epsilon)|Y|}\|(B_Y)_{ij}\|_r \leq Cr. \tag{41}$$

Proof. For $b \in Y = \{i\} \cup X$, one has $d(i,b) \leq \tau(Y)$ and $|Y| \leq |X| + 1$. Thus

$$e^{\mu\tau(Y)+a|Y|} \leq e^a e^{\kappa\tau(Y)+a|X|} e^{-(\kappa-\mu)d(i,b)}. \tag{42}$$

Summing first over X and then i, and using $\sup_b \sum_i e^{-\delta d(i,b)} \leq \sum_{x\in\mathbb{Z}^4} e^{-\delta|x|}$, proves (40) from (33) and (38). This is an anchored sum over all supports containing an arbitrary cell, not only the original output root.

Use the action estimate on radius $2r$ and take derivatives on radius r. Cauchy and polarization give each cell block the dimension-free bound

$$\|(D_i D_j W_Y)\|_r \leq \frac{2e^2}{r^2}\|W_Y\|_{2r}. \tag{43}$$

The row sum contributes at most $|Y|$, absorbed by $\sup_{n\geq 1} ne^{-\epsilon n} < \infty$ in the reserve. This proves (41) and justifies termwise differentiation. Scalar holomorphy gives symmetry of each coefficient. Rows, columns and background support are all in the same Y by construction. □

Taking $\mu \geq 2(\kappa_{\mathrm{G}} + 2m_{\mathrm{G}})$ converts (41) into the strict precision-support norm with metric-tree weight $e^{(\kappa_{\mathrm{G}}+2m_{\mathrm{G}})\ell(Y)}$ used by the Gaussian density algebra. The Gaussian exponents can be chosen positive and sufficiently small compared with the available flat decay. The smallness of the precision norm is then obtained by shrinking the background radius, not by an unsupported claim about a large original source domain.

2.4 The actual current pivot chart and retained precision

Use the fixed-L ordinary pivot and coordinate maps already identified for the physical current step. With $a = \log V/i$, set

$$F_{\mathrm{cur}}(a,z) = \frac{1}{i}\log\left(e^{i\Phi_V(z)}e^{ia}\right), \qquad F_{\mathrm{cur}}(a,0) = a. \tag{44}$$

They are holomorphic on a fixed joint product neighborhood, have a fixed current-cell dependency radius R_c, and satisfy

$$\|F_{\mathrm{cur}}(a,z)\|_\infty \leq C_L(\|a\|_\infty + \|z\|_\infty). \tag{45}$$

Let $G_{\mathrm{cur}}(a,z)$ be the actual local exponential gauge action on $e^{i\Phi_V(z)}V$, not a replacement relative quadratic penalty. Its full analytic expression is finite range at fixed L. At the physical rooted-contour background its quadratic form is the matched gauge term; away from that locus its actual local curvature terms remain included. Put

$$T_M(a,z) = S_M(F_{\mathrm{cur}}(a,z)) + G_{\mathrm{cur}}(a,z). \tag{46}$$

The scalar family for S_M includes its quadratic baseline: split $\frac{1}{2}\langle B, qB\rangle$ into pair-supported scalar coefficients. Exponential locality of q gives their anchored support norm of order r^2 with any fixed cardinality reserve and a sufficiently small positive spatial exponent. Adding (38) therefore gives a scalar support family for the entire S_M.

For a support Y let ΓY be the union of the radius-R_c dependency neighborhoods of its cells. Fixed constants N_c, c_c obey

$$|\Gamma Y| \le N_c|Y|, \qquad \tau(\Gamma Y) \le \tau(Y) + c_c|Y|. \tag{47}$$

Attach a fixed finite neighborhood tree at each point of a minimizing connected graph for Y to prove the second inequality. Periodic identifications only reduce this count. Hence

$$e^{\sigma\tau(\Gamma Y)+a_{\mathrm{out}}|\Gamma Y|} \le e^{\sigma\tau(Y)+(\sigma c_c+a_{\mathrm{out}}N_c)|Y|}. \tag{48}$$

Choose the input cardinality reserve to pay this fixed cost, and shrink the joint radius so that (45) remains in the source-action ball. Substitute the chart in each scalar coefficient and assign support $X = \Gamma Y$. A root in ΓY can be moved to a point of Y in at most a fixed number of ways. Adding the local gauge coefficients gives the exact expansion

$$T_M(a,z) = \sum_X T_X(a|_X, z|_X), \tag{49}$$

with a uniform anchored scalar norm on the fixed joint product domain and any prescribed fixed output cardinality reserve. Both the original background and retained-coordinate labels remain explicit in X.

Corollary 2.4 (Strict support of the physical retained precision)**.** *Define*

$$A_X(a|_X) = D_z^2 T_X(a|_X, 0) - D_z^2 T_X(0,0). \tag{50}$$

These are individually complex-symmetric retained matrices with rows, columns and background dependence in X. For some $\sigma > 0$ and all sufficiently small b,

$$A_M(V) = A_M(1) + \sum_X A_X((\log V/i)|_X), \tag{51}$$

$$\sup_i \sum_{X\ni i}\sum_{j\in X} e^{\sigma\tau(X)}\|(A_X)_{ij}\|_b \le C_L b. \tag{52}$$

The baseline is exactly the matched flat retained precision $A_M(1) = K_L^T(q_M + T_L^T T_L)K_L$.

Proof. Apply Cauchy with fixed margins to the joint domain of (49), twice in z and once in a. Integrating the last derivative along ta gives $O(b)$ in each coefficient. Its row sum costs $|X|$, paid by an available cardinality reserve as in (41). The baseline sum is the flat Hessian of the exact pulled action and hence the identified retained precision. The matrices are Hessians of scalar coefficients, which proves their individual symmetry and supports. □

This argument retains both nonlinear constraint curvature and exponential-coordinate curvature. It does not identify a freely chosen independent source current with the physical Hessian. The full product background extension is an analytic Hessian family; only the physical background is asserted to be stationary. Choosing $2(\kappa_{\mathrm{G}} + 2m_{\mathrm{G}}) < \sigma$ and then shrinking b makes the strict norm as small as the stated Gaussian support algebra requires. The baseline covariance is background independent and has the uniform spectrum and decay proved in the flat theory.

2.5 Localized cubic interaction and Gaussian-cell regrouping

For $\|z\|_\infty \le R$ and gR inside a fixed analytic margin define

$$U_X(a;z) = -g^{-2}\big[T_X(a,gz) - T_X(a,0) - gD_zT_X(a,0)[z] \\ - \tfrac{1}{2}g^2D_z^2T_X(a,0)[z,z]\big]. \tag{53}$$

The sign corresponds to $\exp(-g^{-2}T_M)$. Taylor's integral remainder, or one-variable Cauchy along z/R, yields

$$\sup_i \sum_{X\ni i} e^{\sigma\tau(X)+a_{\mathrm{out}}|X|} \sup_{\|a\|<b,\,\|z\|\le R} |U_X(a;z)| \le CgR^3. \tag{54}$$

No factor $|X|^3$ is needed: the full direction $z|_X$ has product norm at most R. At a physical stationary background the sum of the linear terms in (53) vanishes. This cancellation is not asserted for every point of the analytic product-background domain.

For the fixed retained product cutoff set $\chi_X = \prod_{i\in X}\chi_i$ and $w_X = \chi_X(e^{U_X}-1)$. Then $v_X := \sup|w_X| \le e^{\|U_X\|}\|U_X\|$. Take $a_{\mathrm{out}} > 4 + \log\widehat{K}$ with a positive spare, where $\widehat{K}$ is the chosen fixed Gaussian density factor, and choose $2\kappa_{\mathrm{G}} \le \sigma$. Equations (26) and (54) give

$$\sup_i \sum_{X\ni i} v_X\widehat{K}^{|X|}e^{4|X|+\kappa_{\mathrm{G}}\ell(X)} \le Ce^{CgR^3}gR^3. \tag{55}$$

Thus this is the required localized interaction norm, not only an extensive unweighted cubic remainder. A logarithmically growing R as $g\downarrow 0$ is compatible with both the analytic restriction on gR and smallness of gR^3. The separate Gaussian bad-block tail and the explicit density-algebra smallness conditions must still be imposed. No all-bond nonlinear sharp cutoff is identified with the retained product window by this argument.

The support algebra above uses unit current cells. The Gaussian theorem may use one cell per fixed L-block, with $D = d(L^d-1)\dim\mathfrak{g}$ retained real coordinates in the usual pivot convention. Let π map a unit current cell to its L-block and regroup each support Y to $\pi(Y)$. Then

$$\tau_{\mathrm{block}}(\pi(Y)) \le \tau_{\mathrm{unit}}(Y), \qquad |\pi(Y)| \le |Y|. \tag{56}$$

Every unit edge either stays in one block or crosses to an adjacent block. A block-anchored support sum is bounded by the sum of the unit anchored sums over its at most L^d cells. For matrices, bound a block matrix norm by the sum of its constituent unit-cell blocks, then sum the at most L^d row cells. Thus scalar and absolute-row support norms change by only fixed L-dependent factors. Maximum-cell and Euclidean block norms likewise differ by fixed factors; reduce a background or cutoff radius accordingly. No dimension depending on M is introduced. The Gaussian density constants in (55) can therefore use the fixed retained dimension D after this explicit regrouping.

2.6 Scope of the classical input

The construction expands about a fixed flat baseline in the original current variables. The arbitrary finite cardinality reserve is paid by a smaller uniform product radius; spatial decay alone would not pay the Gaussian factor $\widehat{K}^{|X|}e^{4|X|}$. Fine dimension is removed by complete cell norms, the physical measure, and the actual physical-adjoint row bound. The scalar ordinary action is identified by the proved gauge transfer, while its current pivot chart has fixed ratio L.

The conclusions supply strict-support precision and classical interaction inputs for the homogeneous classical step at every depth M, on the indicated small branch. They do not establish closure of inherited quantum polymer terms, arbitrary independent-current or mixed-history domains, a running-coupling theorem, or an interacting continuum limit. Finite compatible periodic regulators are primary; the

graph estimates are uniform, but source chart incidence still uses the admissible current tori assumed in its proof. No thermodynamic consistency or separate gauge invariance of an arbitrarily regrouped individual coefficient is asserted.

3 One integration of the classical action at every earlier depth

The support theorem upgrades the bare calculation of Notebook 20. Its exact scope matters: S_M is the classical action obtained by minimizing the original Wilson action through the earlier ratio M. The following integral uses S_M and the source gauge and measure factors. It has no previously generated quantum correction E_k. Thus the result is uniform through classical depth, not a claim that the quantum induction is already closed.

Let Φ_V be Notebook 19's exact retained chart, let $a = \log V/i$, and put

$$\mathcal{T}_{M,V}(x) = S_M\big(\log(e^{i\Phi_V(x)}V)/i\big) + \mathcal{G}(e^{i\Phi_V(x)}V). \tag{57}$$

The full product-domain Hessian is $A_M(V) = D_x^2\mathcal{T}_{M,V}(0)$, including the absolute gauge function away from its zero slice. Its flat baseline is $A_{M,0} = K_L^T(q_M + T_L^T T_L)K_L$. The preceding support construction gives local scalar coefficients $\mathcal{T}_{M,Y}$ and symmetric precision coefficients with all their retained rows, columns and background dependence in Y.

Group unit current cells into fixed L-blocks, assigning their retained positive bonds to them. The Gaussian cell dimension is then $D = d(L^d - 1)\dim\mathfrak{g}$, independent of M and volume. Any bounded change of support convention and anchor multiplicity in this grouping costs constants depending only on L. After the spatial-exponent and cardinality reserves proved above, the resulting coefficients have exactly the weighted support norms of Notebook 18.

Define the logarithmic interaction $v_{M,Y}$ by subtracting the constant, linear and quadratic terms from each $-g^{-2}\mathcal{T}_{M,Y}(a, gz)$ and adding the normalized straightening-Jacobian and Haar logarithms of Notebook 19. The gauge Taylor remainder is already included in $\mathcal{T}_{M,Y}$ and is not counted twice. There is a common small product background ball such that, for every fixed required support-cardinality weight and a sufficiently small spatial exponent,

$$\begin{aligned} \nu_M(b) &\le C_L b, \\ \sup_i \sum_{Y\ni i} e^{a_0|Y|+\kappa\ell(Y)}\|v_{M,Y}\|_{b,R} &\le C_{L,a_0,\kappa} g R^3, \qquad R \ge 1, \quad gR < r_L. \end{aligned} \tag{58}$$

Here ν_M is the precision support norm, not just the norm of the summed matrix, and ℓ is the metric tree length used by Notebook 18. All constants are uniform in the earlier classical depth and periodic volume.

Theorem 3.1 (Uniform classical-reference integration)**.** *Fix odd L, the gauge group, $c > 0$ and $p > 1/2$, and set $R(g) = c(\log g^{-2})^p$. There are $b, g_0, a_*, C, c_* > 0$, independent of $M = L^k$ and compatible periodic volume, for which*

$$Z_{M,g}(V) = \int d\mu_{A_M(V)^{-1}}(z)\,\chi_{R(g)}(z)\exp\left(\sum_Y v_{M,Y}(V,z)\right) \tag{59}$$

has a holomorphic logarithm on the independent current background product ball $\|a\|_\infty < b$, for $0 < g < g_0$. There is a convergent connected-support expansion

$$\log Z_{M,g}(V) = \sum_H E_{M,g,H}(V|_H), \qquad \sup_i \sum_{H\ni i} e^{a_*|H|}\|E_{M,g,H}\|_b \le C\{gR(g)^3 + e^{-c_* R(g)^2}\}. \tag{60}$$

Its canonical difference from the pure Gaussian cutoff expansion has norm at most $CgR(g)^3$.

Proof. Notebook 19 supplies the flat spectral constants and weighted inverse uniformly in M. Choose the spatial exponents in Notebook 18 smaller than both that inverse exponent and half the available Steiner exponent for the support coefficients. Choose a fixed $0 < \theta \le 1/4$, for example $1/(4D)$. The arbitrary fixed cardinality reserve in the support theorem pays $4 + \log \mathcal{K}_\theta$ in its interaction norm. The first bound in (58) permits one M-independent $b > 0$ satisfying every local-precision smallness condition of its tunable strictly local theorem.

The interaction parameter is at most CgR^3 by the second bound in (58). The bad-cell parameter is at most $Ce^{-c_* R^2}$ by the correlated real-baseline comparison. The retained-bond rectangles are covered by the cell tail argument of Notebook 20 and remain fixed under the exact chart. A common g_0 makes both quarter budgets hold and keeps the recovered whole field within its analytic radius. Apply Notebook 18 and its periodic extension. □

At a physical background $V_*(W)$, the sum of the separated linear terms is exactly zero and the gauge constant is zero. Thus (59) is the actual fluctuation factor of the constrained classical-reference integral in that chart. On the larger product domain it is the explicitly centered analytic extension, as in Notebook 20.

The normalization factors are equally controlled. With $B_M(V) = A_M(V) - A_{M,0}$ and $C_{M,0} = A_{M,0}^{-1}$, set

$$\mathcal{Q}_M(V) = -\tfrac{1}{2} \log \det(A_M(V) A_{M,0}^{-1}) = \tfrac{1}{2} \sum_{n \ge 1} \frac{(-1)^n}{n} \operatorname{Tr}[(C_{M,0} B_M(V))^n]. \tag{61}$$

The closed-chain proof of Notebook 20 uses only weighted covariance row sums and the precision-label norm. Equation (58) supplies those uniformly in M, even though the precision supports are no longer bounded in size. Joining each label by a minimal tree and each covariance link by a shortest path gives the same bound. Consequently $\mathcal{Q}_M$ has strictly local connected activities with anchored norm at most Cb. The separate delta factor $\log[\mathcal{D}(V)/\mathcal{D}(1)] = -\sum_c \operatorname{Tr} \log \pi_c(V)$ has fixed-range anchored norm at most Cb. Neither normalization term is silently counted as $O(g)$.

For real physical backgrounds define $I^{\mathrm{sf}}_{M,g}(V)$ to be the constrained current group integral with weight $\exp[-g^{-2}(S_M(\log U/i) + \mathcal{G}(U))]$, restricted to the local retained chart and its specified rectangle. Using the notation $N_b, N_c, n, c_H, \mathcal{D}$ from Notebook 20, its exact formula is

$$\begin{aligned} I^{\mathrm{sf}}_{M,g}(V) = {} & c_H^{N_b - N_c} g^n \mathcal{D}(1) (2\pi)^{n/2} (\det A_{M,0})^{-1/2} e^{-S_M(a)/g^2} \\ & \times \exp\left\{ \log \frac{\mathcal{D}(V)}{\mathcal{D}(1)} + \mathcal{Q}_M(V) + \sum_H E_{M,g,H}(V|_H) \right\}. \end{aligned} \tag{62}$$

There is no unaccounted field-dependent normalization in this defined integral. A gauge-insertion identity outside the chart would additionally require S_M itself to be defined and gauge invariant on the chosen admissible rooted-gauge-invariant domain. The small-potential construction does not assert that extension. The exact chart formula above does not require it.

4 The next induction problem

The classical-reference integral is now controlled uniformly over the number of earlier classical averaging steps. This removes a separate localization premise from the homogeneous small-field argument. Its support statement initially records dependence on independent current potentials. Notebook 22 supplies the further strict support expansion of the actual stationary background $V_*(W)$ and transports the generated activities into W, with explicit radius and support-weight losses. Notebook 23 proves local gauge invariance of the complete correction, including the normalization factors, and its quadratic Ward identity.

Notebook 24 then supplies a conditional inherited-interaction bound, and Notebook 26 meets its support-weight test by using a persistent auxiliary-cell convention. The next problem is stable analytic domains and control of the accumulated interaction sizes, with extraction of vacuum and marginal terms and estimates along the required coupling trajectory. The bounded one-step estimates do not supply this recurrence. Notebook 25's smaller representative gives one analytic continuation tool for addressing the domain problem. Large-field complements and mixed histories must be included in the same induction. The continuum observable bounds, symmetry restoration and reconstruction remain the final existence obligations recorded in Notebook 01.

References

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